Electricity

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NCERT Class 10 Science - Chapter 11 (Hinglish Detailed Notes)

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COLOUR CODE: RED = Core Basics / extra help (Class 7-8-9 wali cheezein jo yaad honi chahiye).

COLOUR CODE: GREEN = Board exam me ye topics BAAR-BAAR aate hain - pakka yaad karo.

Baaki normal black text = main chapter content (detail me, simple bhasha me).

CORE BASICS - PEHLE YE PADHO (CLASS 7-8-9 KA FOUNDATION)

Tune 7-8-9 nahi padhi - koi tension nahi. Ye page is chapter ki base hai.
Numerical-heavy chapter hai, isliye base + formula rearrange karna pakka aana chahiye.

1) ELECTRIC CHARGE kya hai?
- Har cheez atom se bani; atom me electron (-ve) aur proton (+ve) hote.
- ELECTRON ke paas chhota -ve charge hota. Charge ka SI unit = COULOMB (C).
- 1 electron ka charge = 1.6 x 10^-19 C (yaani 1 C me ~6.25 x 10^18 electron).
- Charge ka behna (flow) hi CURRENT kehlata hai.

2) CONDUCTOR vs INSULATOR:
- CONDUCTOR = current paas hone de (metal: Cu, Al; free electron hote).
- INSULATOR = current na de (rubber, plastic, wood, glass).

3) CELL aur BATTERY:
- CELL = ek device jo chemical energy ko electric energy me badle; do terminal hote - bada/lamba = +ve, chhota = -ve.
- BATTERY = do ya zyada cell ka combination (zyada voltage ke liye).

4) CIRCUIT kya hai?
- CIRCUIT = ek BAND (closed) raasta jisme current ghoom ke wapas aaye.
- Switch ON = circuit closed (current chale); switch OFF = open (na chale).

5) COMMON CIRCUIT SYMBOLS (yaad rakho - diagram me kaam aayenge):
- Cell -> ek lamba patli line (+) aur ek chhoti moti line (-).
- Battery -> do ya zyada cell side by side.
- Bulb -> circle ke andar cross (X) ya loop (filament).
- Switch/Key -> ek gap with ek chhoti line (open/close).
- Wire/Conductor -> seedhi line.
- Resistor -> ek rectangle (box) ya zig-zag line.
- Ammeter -> circle ke andar letter A.
- Voltmeter -> circle ke andar letter V.

6) CURRENT aur VOLTAGE simple words me:
- CURRENT (I) = wire me charge kitni tezi se beh raha (paani ke flow jaisa).
- VOLTAGE/POTENTIAL DIFFERENCE (V) = charge ko dhakelne wala "push" (paani ke pipe me pressure jaisa). Cell ye push deti hai.

7) SERIES vs PARALLEL (intro):
- SERIES = sab components ek hi line/raaste me ek ke baad ek jude.
- PARALLEL = components alag-alag branch me jude (ek hi do point ke beech).

8) UNITS TABLE (is chapter ke main units):
- Current (I) -> AMPERE (A)
- Charge (Q) -> COULOMB (C)
- Potential diff (V) -> VOLT (V)
- Resistance (R) -> OHM (ohm)
- Resistivity (rho) -> ohm*m (ohm metre)
- Power (P) -> WATT (W)

- Energy -> JOULE (J) ya kWh (kilowatt-hour)

9) FORMULA REARRANGE karna (basic algebra - bahut zaroori):
- Agar V = I R hai to: I = V / R aur R = V / I.
- Yaani jis cheez ko nikalna ho, use akela karo, baaki cheezein dusri taraf.
- Triangle trick: V upar, (I R) niche -> ungli se dhako jo chahiye.
- Units bhi saath le ke chalo (V = volt, A = ampere, ohm = ohm).

1. ELECTRIC CURRENT (VIDYUT DHARA)

DEFINITION: Electric current = kisi conductor ke cross-section se per second behne wale charge ki maatra. Yaani current = charge / time.

FORMULA: $I = Q / t$

- I = current (ampere, A), Q = charge (coulomb, C), t = time (second, s).
- Isse: $Q = I \times t$ aur $t = Q / I$.

UNIT: AMPERE (A). 1 A = 1 C / s

- Matlab agar 1 second me 1 coulomb charge beh raha to current = 1 ampere.
- Chhoti current ke liye: 1 mA (milliampere) = 10^-3 A,
  1 microampere = 10^-6 A.

CURRENT KI DIRECTION (exam favourite):

- CONVENTIONAL current ki direction = +ve charge ke behne ki direction (yaani +ve terminal se -ve terminal taraf bahar circuit me).
- ELECTRON asal me -ve terminal se +ve terminal jaate (ulti direction).
- To conventional current, electron flow ke ULT (opposite) maana jaata.

MAAPNA (measure): current AMMETER se naapte.

- Ammeter ko hamesha SERIES me jodte (jis component me current naapni ho).
- Ideal ammeter ka resistance ZERO (taaki current par asar na pade).

2. ELECTRIC POTENTIAL & POTENTIAL DIFFERENCE

POTENTIAL DIFFERENCE (V) = do point ke beech unit charge ko le jaane me kiya gaya KAAM (work). Yahi charge ko circuit me dhakelta hai.

FORMULA: $V = W / Q$

- V = potential difference (volt, V), W = work/energy (joule, J), Q = charge (coulomb, C).
- Isse: $W = V \times Q$ aur $Q = W / V$.

UNIT: VOLT (V). $1\ V = 1\ J / C$

- Matlab 1 coulomb charge ko le jaane me 1 joule kaam lage to V = 1 volt.

MAAPNA (measure): potential difference VOLTMETER se naapte.

- Voltmeter ko hamesha PARALLEL me jodte (jin do point ke beech V naapni ho).
- Ideal voltmeter ka resistance INFINITE (bahut bada) hota.

Yaad rakhne ki trick:

- Ammeter -> A -> "A" se "series" (current naapta, series me).
- Voltmeter -> Parallel me (potential difference naapta).

3. CIRCUIT DIAGRAM & SYMBOLS

- Circuit diagram = circuit ko symbols se draw karna (asli device ke bajaye).
- Faayda: samajhna aur banana aasaan, sabhi same symbol use karte.

COMMON SYMBOLS (revision):
- Cell / Battery, Bulb, Switch (key), Wire, Resistor (box/zig-zag),
  Ammeter (A in circle, series), Voltmeter (V in circle, parallel),
  Variable resistor / Rheostat (resistor with arrow).

- Ek simple circuit: Cell -> switch -> bulb/resistor -> wapas cell.
  Ammeter series me, voltmeter resistor ke parallel me.

4. OHM'S LAW (BAHUT IMPORTANT)

OHM'S LAW: Constant temperature par, ek conductor me behne wali current (I) uske beech ke potential difference (V) ke DIRECTLY PROPORTIONAL hoti hai.

V proportional to I => V = I R

FORMULA: V = I R

- V = potential difference (V), I = current (A), R = resistance (ohm).
- Isse: I = V / R aur R = V / I.

V-I GRAPH:

- V ko y-axis, I ko x-axis par lo -> graph ek SEEDHI LINE (straight line) origin se hoti hui aati hai.
- Line ka SLOPE = V / I = R (resistance). Zyada slope -> zyada resistance.

RESISTANCE (R):

- R = V / I. Ye conductor ka current ke flow ke against "rukaawat" (opposition).
- UNIT: OHM (likhte: ohm). 1 ohm = 1 volt / 1 ampere.
- 1 ohm = jab 1 V lagane par 1 A current behti.

Note: jo conductor Ohm's law follow kare = OHMIC (e.g. metal wire).
Jo na follow kare (graph straight line na ho) = NON-OHMIC (e.g. bulb, diode).

5. RESISTANCE & FACTORS (RESISTIVITY)

Conductor ka resistance (R) in cheezon par depend karta hai:

(i) LENGTH (L): R, length ke DIRECTLY proportional. Lamba taar -> zyada R.
(ii) AREA (A): R, cross-section area ke INVERSELY proportional.
Mota taar (zyada area) -> kam R.
(iii) MATERIAL: har material ka apna RESISTIVITY (rho) hota.
(iv) TEMPERATURE: temperature badhne par (metal me) R badh jaata.

FORMULA: R = rho * L / A

- rho (rho) = resistivity, L = length (m), A = area (m^2).
- Isse: rho = R * A / L.

RESISTIVITY (rho):

- DEFINITION: rho = us material ke 1 m lambe aur 1 m^2 area wale tukde ka resistance. Ye material ki apni property hai (size par depend nahi karti).
- UNIT: ohm*m (ohm metre).

CONDUCTOR vs INSULATOR by resistivity:

- CONDUCTOR (metal): bahut KAM resistivity (10^-8 ohm*m range).
- INSULATOR (rubber, glass): bahut ZYADA resistivity (10^12 ohm*m range).
- ALLOY (nichrome, manganin): conductor se zyada resistivity + heat me jaldi oxidise nahi hote -> isiliye heating elements me use hote.

Note: silver sabse achha conductor (sabse kam resistivity), uske baad copper.
Ghar ki wiring me COPPER / ALUMINIUM use hota (kam resistivity, saste).

6. RESISTORS IN SERIES

SERIES me resistors ek ke baad ek (single line) jude hote.

KEY POINTS:

- Har resistor me SAME current (I) behti.
- Total voltage = har resistor ke voltage ka JOD (sum):
  $V = V1 + V2 + V3$.
- EQUIVALENT (total) RESISTANCE: $Rs = R1 + R2 + R3$
- Series me total resistance hamesha sabse BADE resistor se bhi ZYADA hoti.

Drawback: ek component fuse/kharab ho to poora circuit band (purani jhaalar ki tarah - ek bulb gaya to sab band). Isliye ghar me series use nahi karte.

7. RESISTORS IN PARALLEL

PARALLEL me sab resistors ek hi do point ke beech alag-alag branch me jude.

KEY POINTS:

- Har resistor par SAME voltage (V) lagta.
- Total current = har branch ki current ka JOD: $I = I1 + I2 + I3$.
- EQUIVALENT RESISTANCE: $1 / Rp = 1/R1 + 1/R2 + 1/R3$
- Parallel me total resistance hamesha sabse CHHOTE resistor se bhi KAM hoti.

GHAR KE APPLIANCES PARALLEL ME KYUN? (exam me pakka aata)

(i) Har appliance ko same full voltage (220 V) milta.
(ii) Ek band ho to baaki chalte rehte (independent on/off).
(iii) Har device apni zaroorat ki current le sakta (total R kam -> zyada current).

8. HEATING EFFECT OF CURRENT (JOULE'S LAW)

Jab current resistor se behti to resistance ki wajah se HEAT paida hoti -
isko current ka HEATING EFFECT kehte.

JOULE'S LAW OF HEATING: $H = I^2 * R * t$

- H = heat (joule, J), I = current (A), R = resistance (ohm), t = time (s).
- Heat directly proportional to: I ke square, R, aur time t.

Why heat? Current behne wale electron, conductor ke ions se takraate, jisse energy heat ke roop me release hoti.

APPLICATIONS (heating effect ka use):

- ELECTRIC HEATER / IRON / TOASTER / GEYSER: nichrome alloy ka coil (high resistivity + high melting point) garam ho jaata.
- ELECTRIC BULB: tungsten FILAMENT (bahut high melting point ~3380 C) garam ho ke chamakta (roshni deta).
- ELECTRIC FUSE (safety device): niche detail me.

ELECTRIC FUSE (bahut important):

- Ek patla taar (tin + lead alloy) jiska melting point KAM hota.
- Circuit ke SERIES me lagaya jaata.
- Jab current safe limit se zyada ho jaye (short circuit / overload), fuse wire $H = I^2 R t$ se garam ho ke PIGHAL jaata -> circuit TOOT jaata.
- Isse appliance aur aag se bachaav hota. Yahi fuse ka kaam hai.

9. ELECTRIC POWER (VIDYUT SHAKTI)

ELECTRIC POWER (P) = per second me use/kharch hone wali electric energy (rate).

FORMULAE (teeno yaad rakho):

$P = V I$ (power = voltage x current)

$P = I^2 * R$ ($V = IR$ daal ke)

$P = V^2 / R$ ($I = V/R$ daal ke)

- P = power (watt, W), V = volt, I = ampere, R = ohm.

UNIT: WATT (W). 1 W = 1 V x 1 A = 1 J/s.

- Bada unit: 1 kilowatt (kW) = 1000 W.

ELECTRIC ENERGY: $E = P \times t$ (power x time).

- SI unit = JOULE (J). Par bijli ka bill JOULE me nahi, bada unit chahiye.

COMMERCIAL UNIT OF ENERGY = KILOWATT-HOUR (kWh) - "1 UNIT" of electricity.

- 1 kWh = 1 kW ka appliance 1 ghante (hour) chalane par use hui energy.
- CONVERSION: 1 kWh = 1000 W x 3600 s = 3.6×10^6 J.

ELECTRICITY BILL - UNITS KAISE NIKALE:

- Units (kWh) = Power(kW) x time(hours used).
- Total units x rate-per-unit = bill amount.
- Tip: Watt ko 1000 se divide karke kW banao; minutes ko 60 se divide karke hours banao.

10. SOLVED EXAMPLES (HARDEST -> EASIEST)

Ye type ke numericals exam me pakka aate - steps + UNITS khud likhke practice karo.

EXAMPLE 1 (Hardest): Combination - 6 ohm aur 3 ohm PARALLEL me hain, aur ye combination 4 ohm ke saath SERIES me. Source = 12 V. Equivalent R aur total current nikaalo.

Step 1 - Parallel part (6 aur 3):

$1/R_p = 1/6 + 1/3 = 1/6 + 2/6 = 3/6 = 1/2$ -> $R_p = 2$ ohm.

Step 2 - Ab Rp (2 ohm) series me 4 ohm ke saath:

R(total) = 2 + 4 = 6 ohm.

Step 3 - Total current (Ohm's law):

I = V / R = 12 / 6 = 2 A.

ANSWER: Equivalent R = 6 ohm, total current = 2 A.

EXAMPLE 2: Parallel - 4 ohm aur 6 ohm parallel me 12 V se jude. Rp aur har branch ki current nikaalo.

Step 1 - Equivalent:

$1/R_p = 1/4 + 1/6 = 3/12 + 2/12 = 5/12$ -> $R_p = 12/5 = 2.4$ ohm.

Step 2 - Parallel me har branch par same V = 12 V:

$I_1 = V/R_1 = 12/4 = 3$ A

$I_2 = V/R_2 = 12/6 = 2$ A

Step 3 - Check total: $I = I_1 + I_2 = 3 + 2 = 5$ A (= 12/2.4, sahi).

ANSWER: Rp = 2.4 ohm, I1 = 3 A, I2 = 2 A.

EXAMPLE 3: Power & bill - ek 1000 W (1 kW) heater roz 2 ghante, 30 din chalta. Rate = Rs 5 per unit. Mahine ka bill nikaalo.

Step 1 - Power = 1000 W = 1 kW.

Step 2 - Time = 2 h/day x 30 = 60 hours.

Step 3 - Energy = P x t = 1 kW x 60 h = 60 kWh (= 60 units).

Step 4 - Bill = 60 x 5 = Rs 300.

ANSWER: 60 units, bill = Rs 300.

EXAMPLE 4: Joule heating - ek 5 ohm resistor me 2 A current 5 minute behti. Kitni heat (H) paida hui?

Step 1 - t ko second me: 5 min = 5 x 60 = 300 s.

Step 2 - H = I^2 * R * t = (2)^2 x 5 x 300 = 4 x 5 x 300 = 6000 J.

ANSWER: H = 6000 J (= 6 kJ).

EXAMPLE 5: Ohm's law - ek bulb par 220 V lagne par 0.5 A current behti. Resistance nikaalo.

R = V / I = 220 / 0.5 = 440 ohm.

ANSWER: R = 440 ohm.

EXAMPLE 6: R = rho*L/A - agar ek taar ki LENGTH double kar di jaye (area same), to resistance par kya asar? Aur agar AREA double ho (length same) to?

- R proportional to L: length double -> R bhi DOUBLE (2 guna).
- R proportional to 1/A: area double -> R HALF (aadha) ho jaata.

ANSWER: length double => R x2 ; area double => R /2.

EXAMPLE 7 (Easiest): I = Q/t - ek conductor se 60 C charge 2 minute me behta.

Current nikaalo.
Step 1 - $t = 2$ min $= 120$ s.
Step 2 - $I = Q / t = 60 / 120 = 0.5$ A.
ANSWER: $I = 0.5$ A.

11. QUICK Q&A - PADHNE KE BAAD KHUD CHECK KARO

Pehle khud answer socho, fir niche milao. Ye board-style questions hain.

Q1. Electric current kya hai aur uska SI unit?
A1. Per second behne wala charge; I = Q/t. SI unit = ampere (A), 1 A = 1 C/s.

Q2. Ammeter aur Voltmeter circuit me kaise jodte hain?
A2. Ammeter SERIES me (current naapne), Voltmeter PARALLEL me (potential difference naapne). Ideal ammeter R = 0, ideal voltmeter R = infinite.

Q3. Ohm's law likho aur V-I graph kaisa hota?
A3. Constant temp par V proportional to I, yaani V = IR. V-I graph origin se nikalti SEEDHI LINE; slope = R.

Q4. Resistance kin cheezon par depend karta? Formula?
A4. Length (proportional L), area (proportional 1/A), material (rho), temperature. R = rho*L/A. Resistivity ka unit = ohm*m.

Q5. Series me equivalent resistance aur ek nuksan?
A5. Rs = R1 + R2 + R3. Nuksan: ek component kharab to poora circuit band.

Q6. Parallel ka formula aur ghar me parallel kyun?
A6. 1/Rp = 1/R1 + 1/R2 + ... Ghar me: har appliance ko full voltage milta aur ek band ho to baaki chalte.

Q7. Joule's law of heating likho aur ek application.
A7. H = I^2 * R * t. Application: electric heater/iron/bulb filament/fuse.

Q8. Electric fuse kaise kaam karta?
A8. Kam melting point ka patla taar series me; zyada current par H = I^2 R t se pighal ke circuit tod deta (overload/short circuit se bachaav).

Q9. Electric power ke teeno formula aur unit?
A9. P = VI = I^2 R = V^2 / R. Unit = watt (W); 1 W = 1 J/s.

Q10. 1 unit bijli = kitna? (kWh aur joule me)
A10. 1 unit = 1 kWh = 1000 W x 3600 s = 3.6 x 10^6 J.

EXAM STRATEGY (1 minute revision):
- Ohm's law V = IR + V-I straight line graph (slope = R) - pakka aata.
- R = rho*L/A + 4 factors (length, area, material, temperature).
- Series Rs = R1+R2+R3 ; Parallel 1/Rp = 1/R1+1/R2+... (signs/steps yaad).
- Joule's law H = I^2 R t + fuse working - scoring topic.
- Power P = VI = I^2 R = V^2/R aur 1 kWh = 3.6 x 10^6 J + bill nikaalna.
- Numericals me UNITS likhna mat bhoolo (marks katte hain warna).
- CORE BASICS page (charge, circuit, symbols, formula rearrange) bhool jaao to wapas pehle padho - base strong rakho.

Generated by Kiro for Krishna1k - All the best!